

Yincheng Zhou

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EDUCATION

University of Pennsylvania

Vagelos Integrated Program in Energy Research (VIPER)

- **Dual Degree:** B.S.E. in Computer Science (AI Concentration) | B.A. in Physics & Astronomy
- **GPA:** 4.0 / 4.0

Philadelphia, PA

September 2025 – Present

EXPERIENCE

GRASP Laboratory | University of Pennsylvania [\[GitHub\]](#)

Philadelphia, PA

Machine Learning Researcher, Advisor: Prof. Pratik Chaudhari

January 2026 – Present

- Proposed SPD, a sparse monocular depth estimator that predicts depth only at queried pixels, achieving $6\times$ inference speedup over dense methods while preserving accuracy (AbsRel 0.11 on NYU Depth V2) for real-time robotics.
- Developed a dense-train / sparse-infer paradigm with a 350K-param split decoder and cross-attention fusion on DINOv2 features, proving sparse output matches dense pixel-wise prediction through systematic ablation across 15 architecture variants.

Welta, Inc. | AI-powered cross-platform ad optimization engine [\[Demo\]](#)

Remote

Software Engineer Intern

March 2026 – Present

- Designed ad spend optimizer using Bayesian response curves and convex optimization to reallocate budget across channels at fixed cost; validated on 93 e-commerce brands, delivering 23% higher return on ad spend versus manual allocation.
- Engineered ad simulation engine that tokenizes creatives and models persona-level reactions via LLM; reduced inference latency to 20% of baseline while improving prediction accuracy by 55%.

Human Machine Intelligence Laboratory | Peking University [\[HuggingFace\]](#)

Remote

Machine Learning Researcher, Advisor: Prof. Shanghang Zhang

March 2026 – Present

- Developed MCTS-guided SFT for GUI agent training that collects and trains on step-level decisions with difficulty weighting, outperforming trajectory-level RL (GRPO) at 25% compute; raised UI-TARS-7B's OSWorld success from 12.3% to 17.7%.
- Engineered distributed rollout infrastructure across 112 Docker VMs and 8 A100 GPUs with fault-tolerant container monitoring and action-replay VM cloning; collected 2,927 trajectories and curated 35K steps via 200 parallel LLM auditing agents.

Franklink, Inc. | World's first AI-native networking platform [\[Demo\]](#)

Philadelphia, PA

Founding Software Engineer

October 2025 – February 2026

- Designed iMessage-based multi-agent orchestration layer with persisted memory, selective dispatch across 5 sub-agents, and agentic RAG over 8 permissioned tools (DB, email, calendar); reduced user-facing latency by 30%.
- Engineered a Kafka-backed event pipeline with at-least-once delivery, idempotency keys, and retry + DLQ; sustained 50+ concurrent sessions with p99 latency within 40% of single-session baseline; deployed on AWS ECS.

PROJECTS

Heard | Anthropic×Penn Hackathon: democratic advocacy platform [\[GitHub\]](#) [\[Demo\]](#)

April 2026

- Champion in Democratic Governance & Collaboration Track, Final 3rd Place among ~400 participants.
- Converted citizen grievances into ready-to-send advocacy and root-cause analysis with elected-official mapping across 4 government levels; two-sided platform pairing one-click citizen outreach with a district-aware official dashboard.

AlphaOne | Subject-oriented financial sentiment platform [\[GitHub\]](#) [\[Demo\]](#)

August 2025 – February 2026

- Fine-tuned DeBERTa for per-ticker aspect-based sentiment in multi-entity financial text, achieving 82.5% accuracy (+65pp over FinBERT); curated 6,287 training triples via Reddit scraping.
- Engineered a 3-pass batch NLP pipeline (split, infer, commit) with claim-based row locking for concurrent-safe ingestion across 7 subreddits; serves real-time per-ticker trends through a full-stack dashboard.

F-Zero Racing Agent | RL agent for competitive SNES racing [\[GitHub\]](#)

February 2026 – April 2026

- Trained a PPO agent across 80 parallel SNES emulators with custom reward shaping to race F-Zero, reaching global top 300 on a task previously deemed unsolvable due to sparse terminal-only rewards over 3,000+ step horizons.
- Shaped dense reward via spline-interpolated track centerline; dual-input CNN+MLP policy fusing visual and state streams.

IncidentGuard | Multi-agent incident remediation system [\[GitHub\]](#)

March 2026 – April 2026

- Simulates outcomes before execution to reject unsafe plans; 46% faster recovery across 4 failure scenarios.
- Closed-loop plan-execute-verify pipeline with Prometheus metrics; graceful degradation across 3 fallback paths.

NL2SQL Bot | Agentic NL-to-SQL orchestration [\[GitHub\]](#)

December 2025 – January 2026

- Orchestrated 5 agents on Google ADK with loop-guarded retry, per-agent model routing, and automated 6-type chart generation.
- Enforced read-only SQL via 18 blocked patterns and table allowlisting; cross-dialect portability across 3 SQL engines.

SKILLS & ACHIEVEMENTS

Languages: C++, Python, Java, JavaScript/TypeScript, SQL, Assembly, OCaml

Backend & Infrastructure: FastAPI, Spring Boot, Kafka, Redis, Celery, Docker, AWS, Ray, Linux, Git

ML & Data: PyTorch, CUDA, vLLM, HuggingFace Transformers, spaCy, Pandas, LangGraph

Achievements: Team China, International Science and Engineering Fair (ISEF); Silver Medal, Team China, International Young Physicists' Tournament (IYPT); Second Prize, China Mathematics Olympiad; Second Prize, China Physics Olympiad